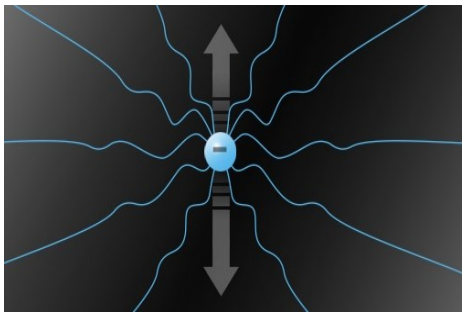


XIAMEN UNIVERSITY MALAYSIA

EP2T #4

Electric charge and electric field

Lecture 4



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Question 1:

Two small spheres spaced 20.0 cm apart have equal charge. How many excess electrons must be present on each sphere if the magnitude of the force of repulsion between them is 3.33×10^{-21} N?

Question 2:

Two small plastic spheres are given positive electric charges. When they are 15.0 cm apart, the repulsive force between them has magnitude 0.220 N. What is the charge on each sphere (a) if the two charges are equal and (b) if one sphere has four times the charge of the other?

Question 3:

A negative charge of $-0.550\ \mu$C exerts an upward 0.600-N force on an unknown charge that is located 0.300 m directly below the first charge. What are (a) the value of the unknown charge (magnitude and sign); (b) the magnitude and direction of the force that the unknown charge exerts on the -0.550-μC charge?

Question 4:

Three point charges are arranged along the x-axis. Charge $q_1 = +3.00\ \mu$C is at the origin, and charge $q_2 = -5.00\ \mu$C is at $x = 0.200$ m. Charge $q_3 = -8.00\ \mu$C. Where is q_3 located if the net force on q_1 is 7.00 N in the $-x$-direction?

Question 5:

Two point charges are placed on the x-axis as follows: Charge $q_1 = +4.00\,\text{nC}$ is located at $x = 0.200\,\text{m}$, and charge $q_2 = +5.00\,\text{nC}$ is at $x = -0.300\,\text{m}$. What are the magnitude and direction of the total force exerted by these two charges on a negative point charge $q_3 = -6.00\,\text{nC}$ that is placed at the origin?

Question 6:

A proton is placed in a uniform electric field of $2.75 \times 10^3\,\text{N/C}$. Calculate (a) the magnitude of the electric force felt by the proton; (b) the proton's acceleration; (c) the proton's speed after $1.00\,\mu\text{s}$ in the field, assuming it starts from rest.

Question 7:

A $+8.75$-μC point charge is glued down on a horizontal frictionless table. It is tied to a -6.50-μC point charge by a light, nonconducting 2.50-cm wire. A uniform electric field of magnitude 1.85×10^8 N/C is directed parallel to the wire, as shown in **Fig. T3.1** . (a) Find the tension in the wire. (b) What would the tension be if both charges were negative?

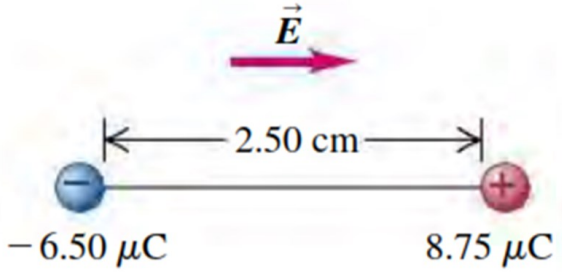


Figure T3.1

Question 8:

Point charge $q_1 = -5.00$ nC is at the origin and point charge $q_2 = +3.00$ nC is on the x-axis at $x = 3.00$ cm. Point P is on the y-axis at $y = 4.00$ cm. (a) Calculate the electric fields $\vec{E}_1$ and $\vec{E}_2$ at point P due to the charges q_1 and q_2. Express your results in terms of unit vectors . (b) Use the results of part (a) to obtain the resultant field at P, expressed in unit vector form.

Question 9:

A point charge $q_1 = -4.00\ \text{nC}$ is at the point $x = 0.600\ \text{m}$, $y = 0.800\ \text{m}$, and a second point charge $q_2 = +6.00\ \text{nC}$ is at the point $x = 0.600\ \text{m}$, $y = 0$. Calculate the magnitude and direction of the net electric field at the origin due to these two point charges.

Question 10:

The dipole moment of the water molecule (H_2O) is $6.17 \times 10^{-30}\ \text{C} \cdot \text{m}$. Consider a water molecule located at the origin whose dipole moment $\vec{\boldsymbol{p}}$ points in the $+x$-direction. A chlorine ion $(C1^-)$, of charge $-1.60 \times 10^{-19}\ \text{C}$, is located at $x = 3.00 \times 10^{-9}\ \text{m}$. Find the magnitude and direction of the electric force that the water molecule exerts on the chlorine ion. Is this force attractive or repulsive? Assume that x is much larger than the separation d between the charges in the dipole.

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